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Create a SQL warehouse

April 11, 2024

Workspace admins and sufficiently privileged users can configure and manage SQL warehouses. This article describes how to create, edit, and monitor existing SQL warehouses.

You can also create SQL warehouses using the [SQL warehouse API](#), or [Terraform](#).

Databricks recommends using serverless SQL warehouses when available.

Note

Most users cannot create SQL warehouses, but can restart any SQL warehouse they can connect to. See [What is a SQL warehouse?](#).

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Requirements

SQL warehouses have the following requirements:

- To create a SQL warehouse you must be a workspace admin or a user with unrestricted cluster creation permissions.
- Before you can create a serverless SQL warehouse in a [region that supports the feature](#), there might be required steps. See [Enable serverless SQL warehouses](#).

Create a SQL warehouse

To create a SQL warehouse using the web UI:

1. Click **SQL Warehouses** in the sidebar.
2. Click **Create SQL Warehouse**.
3. Enter a **Name** for the warehouse.
4. (Optional) Configure warehouse settings. See [Configure SQL warehouse settings](#).
5. (Optional) Configure advanced options. See [Advanced options](#).
6. Click **Create**.
7. (Optional) Configure access to the SQL warehouse. See [Manage a SQL warehouse](#).

Your created warehouse starts automatically.



New SQL warehouse



Name

SQL warehouse name

Cluster size ⓘ

X-Large80 DBU / h

Auto stop

☒

After

10

minutes of inactivity.

Scaling ⓘ

Min.

1

Max.

1

clusters (80 DBU)

Type

☒ Serverless ⓘ

☐ Pro ⓘ

☐ Classic

Serverless SQL warehouses contain all advanced features and are the default warehouse type.

Prices are reduced (up to 20%) until Jan 31, 2024. Try a Serverless SQL warehouse today!
[Learn more](#)

Advanced options >

Cancel

Create

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Configure SQL warehouse settings

You can modify the following settings while creating or editing a SQL warehouse:

- **Cluster Size** represents the size of the driver node and number of worker nodes associated with the cluster. The default is **X-Large**. To reduce query latency, increase the size.
- **Auto Stop** determines whether the warehouse stops if it's idle for the specified number of minutes. Idle SQL warehouses continue to accumulate DBU and cloud instance charges until they are stopped.
 - **Pro and classic SQL warehouses:** The default is 45 minutes, which is recommended for typical use. The minimum is 10 minutes.
 - **Serverless SQL warehouses:** The default is 10 minutes, which is recommended for typical use. The minimum is 1 minute when you use the UI. Note that you can create a serverless SQL warehouse using the [SQL warehouse API](#), in which case you can set the Auto Stop value as low as 1 minute.
- **Scaling** sets the minimum and maximum number of clusters that will be used for a query. The default is 1. You can increase the maximum number of clusters if you want to handle more concurrent queries. Databricks recommends a cluster for every 10 concurrent queries.

To maintain optimal performance, Databricks periodically recycles clusters. During a recycle period, a cluster count that exceeds the maximum as Databricks transitions new workloads to the new cluster will result in the old cluster until all open workloads have completed.
- **Type** determines the type of warehouse. If serverless is enabled in your account, serverless is the default. See [SQL warehouse types](#) for the list.



Advanced options

Configure the following advanced options by expanding the **Advanced options** area when you create a new SQL warehouse or edit an existing SQL warehouse. You can also configure these options using the [SQL Warehouse API](#).

- **Tags:** Tags allow you to monitor the cost of cloud resources used by users and groups in your organization. You specify tags as key-value pairs.
- **Unity Catalog:** If Unity Catalog is enabled for the workspace, it is the default for all new warehouses in the workspace. If Unity Catalog is not enabled for your workspace, you do not see this option. See [What is Unity Catalog?](#).
- **Channel:** Use the Preview channel to test new functionality, including your queries and dashboards, before it becomes the Databricks SQL standard.

The [release notes](#) list what's in the latest preview version.

Important!

Databricks recommends against using a preview version for production workloads. Because only workspace admins can view a warehouse's properties, including its channel, consider indicating that a Databricks SQL warehouse uses a preview version in that warehouse's name to prevent users from using it for production workloads.

Manage a SQL warehouse

Workspace admins and users with CAN MANAGE privileges on a SQL warehouse can complete the following tasks to manage the warehouse:

- To stop a running warehouse, click the stop icon next to the warehouse.
- To start a stopped warehouse, click the start icon next to the warehouse.
- To edit a warehouse, click the kebab menu  then click **Edit**.
- To add and edit permissions, click the kebab menu  then click **Permissions**. To learn about permission levels, see [SQL warehouse ACLs](#).
- To upgrade a SQL warehouse to serverless, click the kebab menu , then click **Upgrade to Serverless**.
- To delete a warehouse, click the kebab menu , then click **Delete**.

Note

Contact your Databricks representative to restore a deleted warehouses within 14 days.



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